

THE IMPACT OF SOCIOECONOMIC STATUS, LIFESTYLE CHOICES, & DEMOGRAPHIC FACTORS ON HEALTH

AN EXPLORATORY DATA ANALYSIS



By DATA350 Group 11: Adrien Rozario, Ademir Ferreyra, and Jay Nadkarni

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INTRODUCTION

Socioeconomic, behavioral, and demographic factors influence health outcomes



Understanding the relationship between these factors can help inform public health interventions

BACKGROUND



NATIONAL
CENTER FOR
HEALTH
STATISTICS



- National Health And Nutrition Examination Survey (2009–2012)
 - 20,293 participants in the U.S.
 - Assessed public health trends, lifestyles, & illness risk factors
 - Accessible through NHANESraw dataset from the NHANES R package
- So how do socioeconomic status, lifestyle choices, and demographic factors impact health?

RESEARCH QUESTIONS

Q1

How does socioeconomic status (education, income, poverty) relate to smoking, alcohol, and drug use?



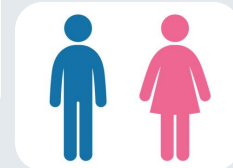
Q2

How do lifestyle choices (smoking, alcohol, and drug use) relate to self-rated health and sleep duration?



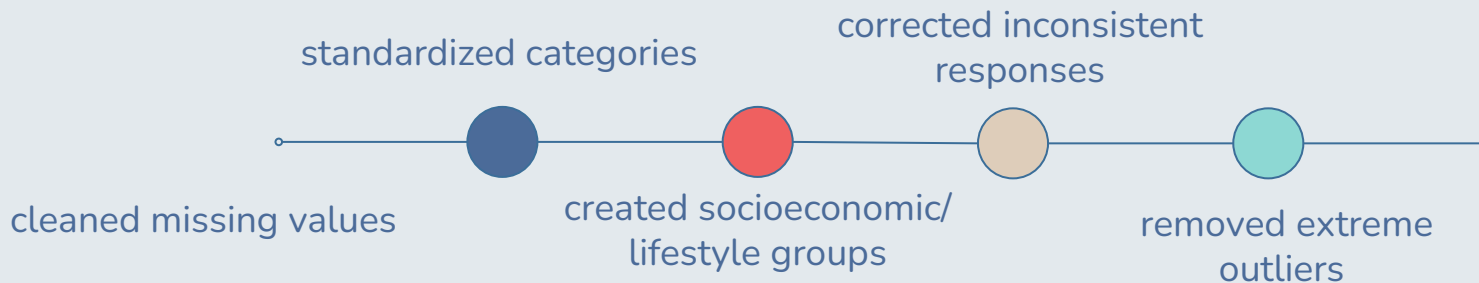
Q3

How do demographic factors (marital status, gender) influence or interact with these lifestyle and health outcomes?



DATA OVERVIEW

Numeric	HHIncomeMid, Poverty, SleepHrsNight, AlcoholDay/Year
Categorical	Education, Gender, Marital Status, SmokeNow, AlcoholGroup, HardDrugs
Ordinal	HealthGen (Excellent, Good, Fair, Poor)



INTERACTIVE DASHBOARD

The Impact of Socioeconomic Status, Lifestyle Choices, & Demographic Factors on Health: An Exploratory Data Analysis Dashboard

Socioeconomic Status & Lifestyle Choices

Lifestyle Choices & Health Outcomes

Demographic Factors & Health Outcomes

Summary

Filters

Gender

- ☒ female
- ☒ male

Education

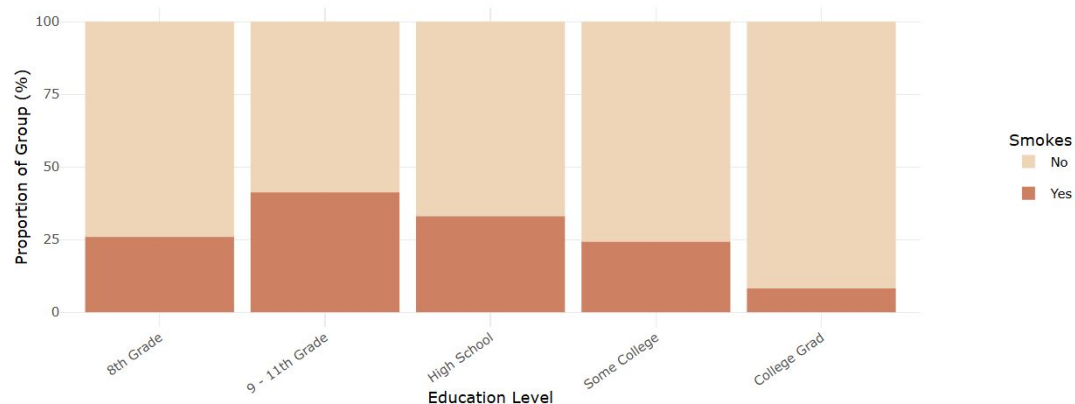
- ☒ 8th Grade
- ☒ 9 - 11th Grade
- ☒ High School
- ☒ Some College
- ☒ College Grad

Poverty Group

- ☒ Low
- ☒ Medium
- ☒ High

Q1: How does socioeconomic status relate to smoking, alcohol, and drug use?

Bar plot of smoking rates by education level



TAB 1

How does socioeconomic status (education, income, poverty) relate to smoking, alcohol, and drug use?

Filters

Gender

- ☒ female
- ☒ male

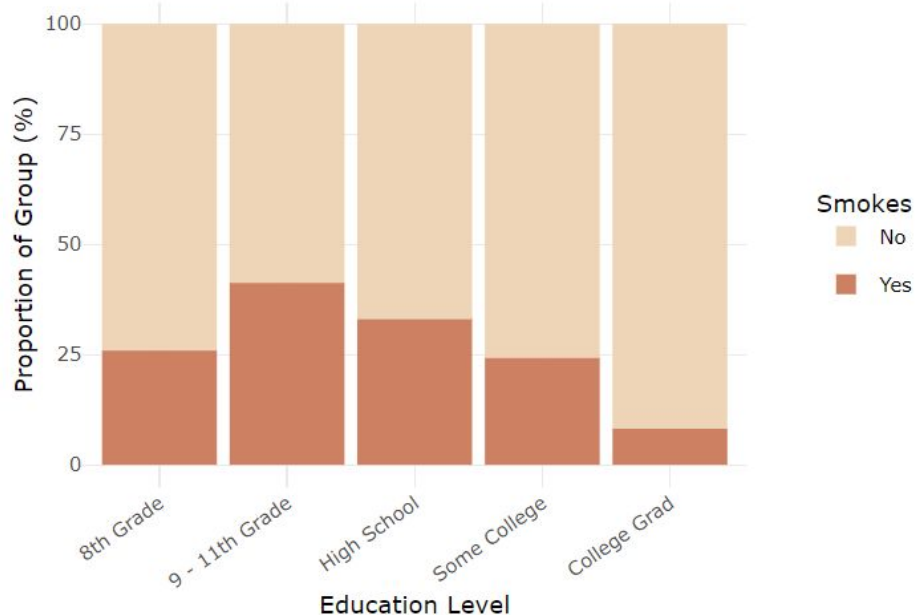
Education

- ☒ 8th Grade
- ☒ 9 - 11th Grade
- ☒ High School
- ☒ Some College
- ☒ College Grad

Poverty Group

- ☒ Low
- ☒ Medium
- ☒ High

Bar plot of smoking rates by education level



TAB 1

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Gender

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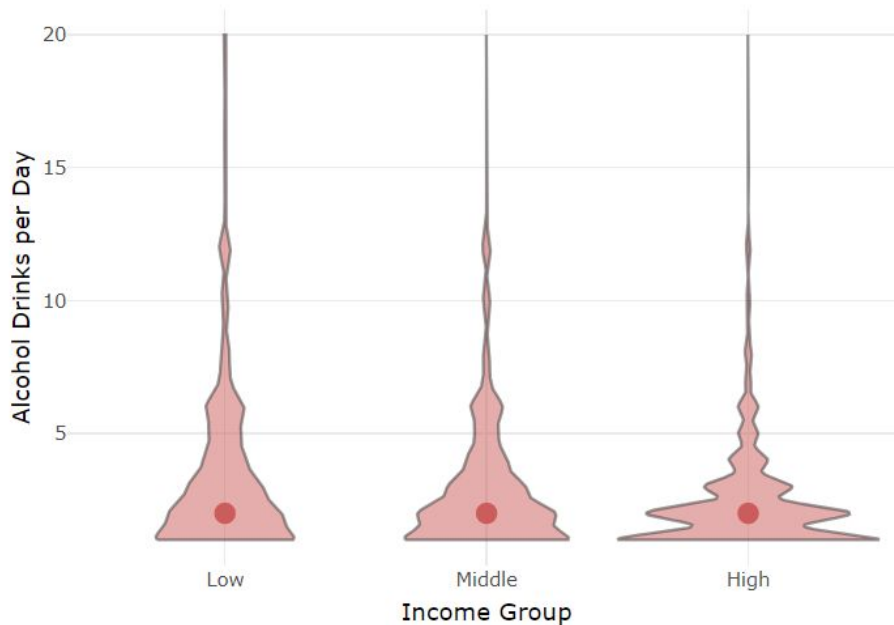
Education

- ☒ 8th Grade
- ☒ 9 - 11th Grade
- ☒ High School
- ☒ Some College
- ☒ College Grad

Poverty Group

- ☒ Low
- ☒ Medium
- ☒ High

Violin plot of alcohol consumption by income group



TAB 1

How does socioeconomic status (education, income, poverty) relate to smoking, alcohol, and drug use?

Filters

Gender

- ☒ female
- ☒ male

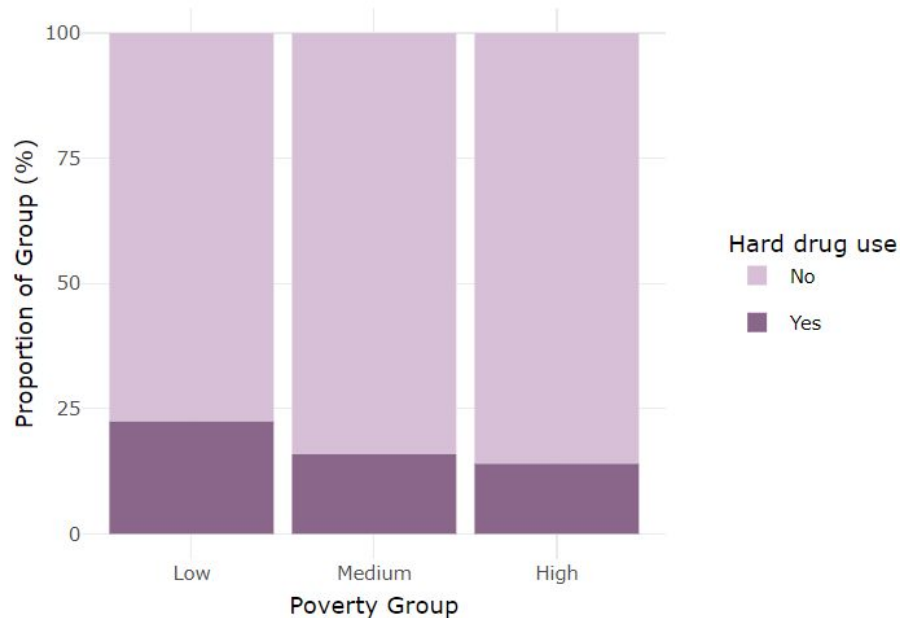
Education

- ☒ 8th Grade
- ☒ 9 - 11th Grade
- ☒ High School
- ☒ Some College
- ☒ College Grad

Poverty Group

- ☒ Low
- ☒ Medium
- ☒ High

Stacked bar plot of hard drug use by poverty level



TAB 2

How do lifestyle choices (smoking, alcohol, and drug use) relate to self-rated health and sleep duration?

Filters

Gender

- ☒ female
- ☒ male

Smoking status

- ☒ No
- ☒ Yes

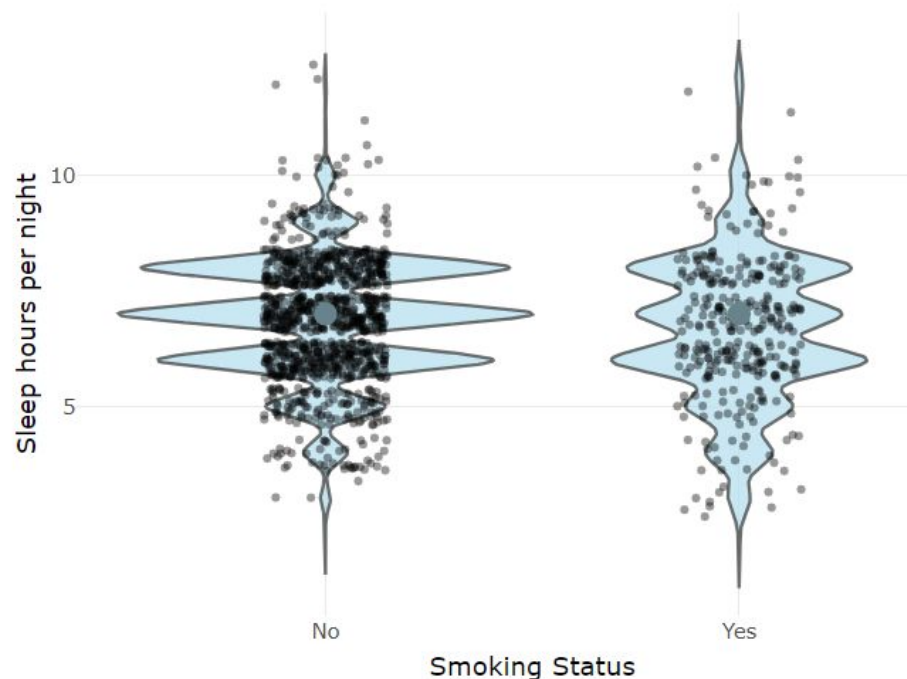
Alcohol level

- ☒ None
- ☒ Light
- ☒ Moderate
- ☒ Heavy

Add smoothing to scatter/jitter?

- ☐ No
- ☒ loess
- ☐ lm

Violin plot of sleep duration by smoking status



TAB 2

How do lifestyle choices (smoking, alcohol, and drug use) relate to self-rated health and sleep duration?

Filters

Gender

- ☒ female
- ☒ male

Smoking status

- ☒ No
- ☒ Yes

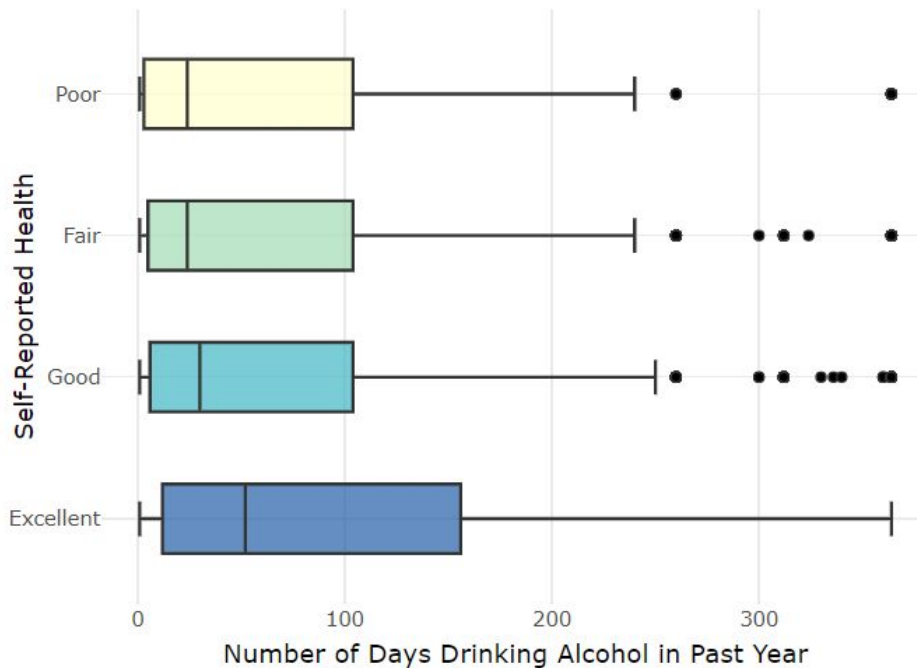
Alcohol level

- ☒ None
- ☒ Light
- ☒ Moderate
- ☒ Heavy

Add smoothing to scatter/jitter?

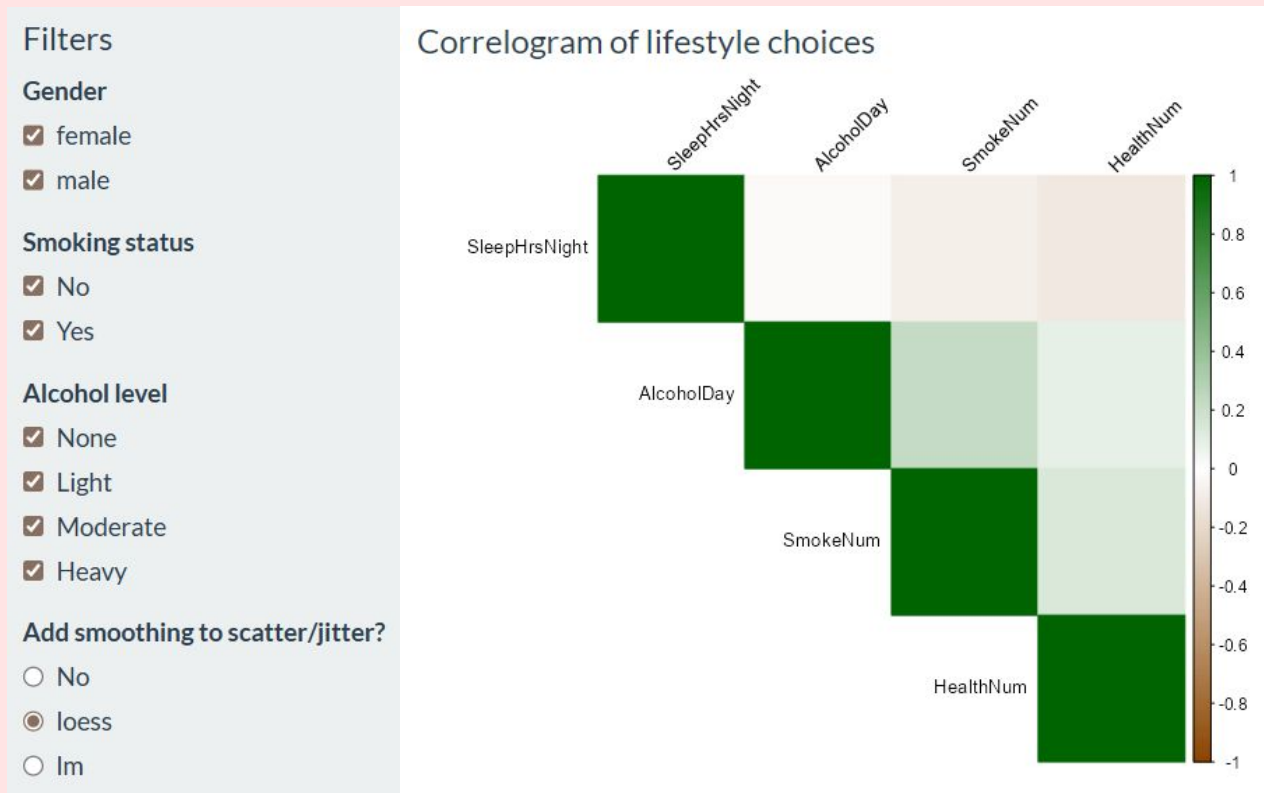
- ☐ No
- ☒ loess
- ☐ lm

Box plot of self-rated health by alcohol consumption



TAB 2

How do lifestyle choices (smoking, alcohol, and drug use) relate to self-rated health and sleep duration?



TAB 3

How do demographic factors (marital status, gender) influence or interact with these lifestyle and health outcomes?

Filters

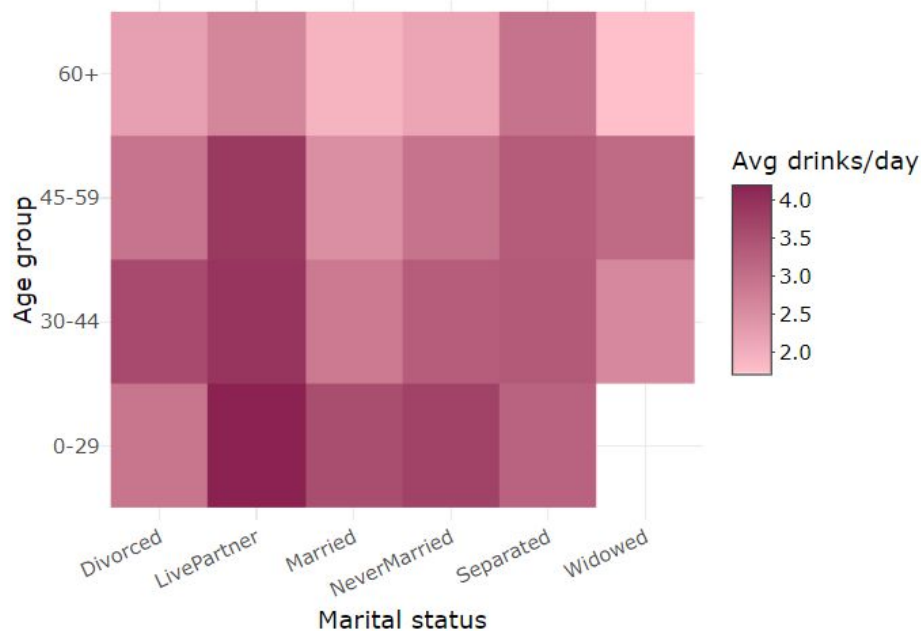
Gender

- ☒ female
- ☒ male

Marital status

- ☒ Married
- ☒ LivePartner
- ☒ NeverMarried
- ☒ Divorced
- ☒ Widowed
- ☒ Separated
- ☒ NA

Heatmap of average daily alcohol use by age group and marital status



TAB 3

How do demographic factors (marital status, gender) influence or interact with these lifestyle and health outcomes?

Filters

Gender

☒ female

☒ male

Marital status

☒ Married

☒ LivePartner

☒ NeverMarried

☒ Divorced

☒ Widowed

☒ Separated

☒ NA

Scatter plot of sleep vs age (faceted by marital status, colored by gender)



SUMMARY TAB

Choose variables (up to 5):

AlcoholYear SmokeNow

Generate summary

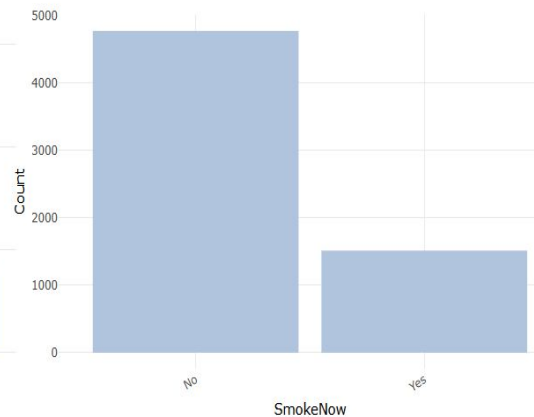
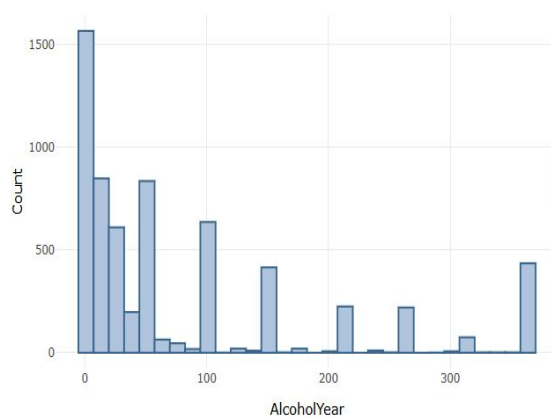
Download full cleaned CSV

Summary output

```
$AlcoholYear
      Min 1st Qu.  Median      Mean 3rd Qu.      Max       SD
1.00000  8.00000  36.00000  83.11773 104.00000 364.00000 105.87200

$SmokeNow
      Count Percent
[1,]  4769    75.9
[2,]  1514    24.1
```

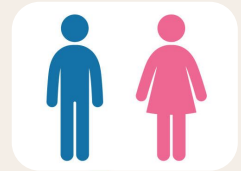
Distribution plots



DASHBOARD DEMO

INSIGHTS

- Lower socioeconomic status is associated with higher smoking & drug use
- Alcohol consumption varies by income but also lifestyle factors
- Sleep and self-rated health weakly influenced by lifestyle
- Demographics add variability (age, marital status)



CONCLUSION

There are limitations:

- Oversampled dataset -> may affect graph displays
- Data from 2009–2012, may not reflect current lifestyle trends
- Sparse data in some subgroups limits conclusions

But overall:

- Complex interplay of SES, lifestyle, demographics -> health outcomes
- Higher SES generally aligns with healthier behaviors
- Visualizations highlight patterns but sample limitations exist
- Future work: more recent datasets, statistical modeling to confirm patterns

THANK YOU

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